Statistics & Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Measures of center**

Find the mean of $12, 15, 18, 9, 21$.

Sum $= 75$. Mean $= \frac{75}{5} = 15$.

2 **[Foundational]** • **Target: Measures of center**

Find the median of $4, 9, 2, 11, 7, 15$.

Sorted: $2, 4, 7, 9, 11, 15$. Median $= \frac{7+9}{2} = 8$.

3 **[Foundational]** • **Target: Measures of center**

Find the mode of $3, 7, 5, 7, 2, 7, 9$.

The value 7 appears most often, so the mode is 7.

4 **[Foundational]** • **Target: Measures of spread**

Find the range of the data set $22, 18, 35, 9, 27$.

Range $= 35 - 9 = 26$.

5 **[Foundational]** • **Target: Measures of spread**

Find the interquartile range (IQR) of $4, 7, 9, 12, 15, 18, 21$.

Lower half (excluding the median 12): $4, 7, 9$, so $Q_1 = 7$. Upper half: $15, 18, 21$, so $Q_3 = 18$. IQR $= 18 - 7 = 11$.

6 **[Foundational]** • **Target: Measures of spread**

A data set has a range of 40 and a minimum value of 15. Find the maximum value.

Maximum $= 15 + 40 = 55$.

7 **[Foundational]** • **Target: Basic probability**

A jar has 5 red, 3 blue, and 2 green marbles. Find $P(\text{green})$.

Total marbles $= 10$. $P(\text{green}) = \frac{2}{10} = \frac{1}{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Basic probability

A fair six-sided die is rolled once. Find $P(\text{even number})$.

Even outcomes: $\{2, 4, 6\}$. $P = \frac{3}{6} = \frac{1}{2}$.

9 [Intermediate] • Target: Basic probability

A spinner has 8 equal sections numbered 1 through 8. Find $P(\text{number greater than } 5)$.

Numbers greater than 5: $\{6, 7, 8\}$. $P = \frac{3}{8}$.

10 [Intermediate] • Target: Combined probability: AND / OR

A card is drawn from a standard 52-card deck. Find $P(\text{king or queen})$.

4 kings $+4$ queens $= 8$ favorable cards. $P = \frac{8}{52} = \frac{2}{13}$.

11 [Intermediate] • Target: Combined probability: AND / OR

Two independent events have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$P(A \cap B) = 0.4 \times 0.5 = 0.20$.

12 [Intermediate] • Target: Combined probability: AND / OR

A bag has 6 yellow and 4 purple marbles. A marble is drawn, replaced, then a second is drawn. Find $P(\text{both yellow})$.

$P = \frac{6}{10} \times \frac{6}{10} = \frac{36}{100} = \frac{9}{25}$.

13 [Intermediate] • Target: Probability without replacement

A bag has 5 red and 7 blue marbles. Two marbles are drawn without replacement. Find $P(\text{both red})$.

$P = \frac{5}{12} \times \frac{4}{11} = \frac{20}{132} = \frac{5}{33}$.

14 [Intermediate] • Target: Probability without replacement

A committee of 2 is chosen at random, one at a time without replacement, from a group of 4 men and 6 women. Find $P(\text{both chosen are women})$.

$P = \frac{6}{10} \times \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Data displays • Cross-Lesson: 6.01 Geometry

The bar graph shows the results of rolling a fair die 40 times. Based on the experimental data, find the experimental probability of rolling a 4, and compare it to the theoretical probability.

Experimental $P(4) = \frac{9}{40} = 0.225$. Theoretical $P(4) = \frac{1}{6} \approx 0.167$. The experimental probability is somewhat higher than the theoretical probability, which is expected random variation in a finite sample.

16 [Challenging] • Target: Expected value • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A game costs 2 dollars to play. A player wins 10 dollars with probability 0.1, and wins nothing otherwise. Find the expected net profit per play.

Expected winnings $= 0.1 \times 10 = 1$ dollar. Expected net profit $= 1 - 2 = -1$ dollar (an expected loss of 1 dollar per play).

17 [Challenging] • Target: Expected value • Cross-Lesson: 8.01 Exponentials & Logarithms

A spinner has 4 equal sections labeled 5, 10, 15, and 20 points. Find the expected value of one spin.

$E = \frac{5 + 10 + 15 + 20}{4} = \frac{50}{4} = 12.5$ points.

18 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 1.01 Heart of Algebra

A data set has mean 20. If every value in the set is increased by 5, find the new mean.

New mean $= 20 + 5 = 25$.

19 [Challenging] • Target: Effects of transformations on statistics • Cross-Lesson: 2.01 Quadratics & Polynomials

A data set has standard deviation 4. If every value is multiplied by 3, find the new standard deviation.

New standard deviation $= 3 \times 4 = 12$.

20 [Challenging] • Target: A well-designed word problem: a survey of students • Cross-Lesson: 3.01 Functions & Graphs

This question has three parts. A researcher surveys 200 students about whether they play a sport and whether they play a musical instrument. Of the 200 students, 120 play a sport, 80 play an instrument, and 30 do both. (a) Find how many students do neither activity. (b) Find the probability that a randomly chosen student plays a sport or an instrument (or both). (c) Given that a randomly selected student plays a sport, find the probability that they also play an instrument.

(a) By inclusion-exclusion, the number doing at least one activity is $120 + 80 - 30 = 170$. Neither $= 200 - 170 = 30$. (b) $P(\text{sport or instrument}) = \frac{170}{200} = 0.85$. (c) $P(\text{instrument} \mid \text{sport}) = \frac{P(\text{both})}{P(\text{sport})} = \frac{30}{120} = \frac{1}{4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Measures of center

A jar has 11 yellow, 6 red, and 6 green marbles. Find $P(\text{red})$.

a. $\frac{6}{23}$ ✓ **b.** $\frac{8}{23}$

c. $\frac{11}{23}$ **d.** $\frac{9}{23}$

Total marbles $= 23$. $P(\text{red}) = \frac{6}{23}$.

22 [Foundational] • Target: Measures of spread

A jar has 3 green, 5 blue, and 12 purple marbles. Find $P(\text{blue})$.

a. $\frac{3}{20}$ **b.** $\frac{7}{20}$

c. $\frac{3}{5}$ **d.** $\frac{1}{4}$ ✓

Total marbles $= 20$. $P(\text{blue}) = \frac{5}{20} = \frac{1}{4}$.

23 [Foundational] • Target: Basic probability

A jar has 6 red, 2 orange, and 2 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{5}$ **b.** $\frac{3}{5}$ ✓

c. $\frac{4}{5}$ **d.** $\frac{9}{10}$

Total marbles $= 10$. $P(\text{red}) = \frac{6}{10} = \frac{3}{5}$.

24 [Foundational] • Target: Combined probability: AND / OR

A jar has 4 red, 8 purple, and 4 green marbles. Find $P(\text{red})$.

a. $\frac{7}{16}$ **b.** $\frac{3}{8}$

c. $\frac{1}{4}$ ✓ **d.** $\frac{1}{2}$

Total marbles $= 16$. $P(\text{red}) = \frac{4}{16} = \frac{1}{4}$.

25 [Foundational] • Target: Probability without replacement

A jar has 2 green, 1 orange, and 10 yellow marbles. Find $P(\text{orange})$.

a. $\frac{1}{13}$ ✓ **b.** $\frac{2}{13}$

c. $\frac{3}{13}$ **d.** $\frac{10}{13}$

Total marbles $= 13$. $P(\text{orange}) = \frac{1}{13}$.

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26 [Foundational] • Target: Data displays

A jar has 9 red, 2 green, and 10 blue marbles. Find $P(\text{red})$.

a. $\frac{2}{21}$

b. $\frac{10}{21}$

c. $\frac{11}{21}$

d. $\frac{3}{7}$ ✓

Total marbles $= 21$. $P(\text{red}) = \frac{9}{21} = \frac{3}{7}$.

27 [Foundational] • Target: Expected value

A jar has 1 yellow, 5 purple, and 5 orange marbles. Find $P(\text{yellow})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$

c. $\frac{3}{11}$

d. $\frac{1}{11}$ ✓

Total marbles $= 11$. $P(\text{yellow}) = \frac{1}{11}$.

28 [Foundational] • Target: Effects of transformations on statistics

A jar has 2 red, 5 purple, and 1 orange marbles. Find $P(\text{red})$.

a. $\frac{1}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{8}$

d. $\frac{1}{4}$ ✓

Total marbles $= 8$. $P(\text{red}) = \frac{2}{8} = \frac{1}{4}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: a survey of students

A jar has 7 purple, 11 orange, and 10 green marbles. Find $P(\text{green})$.

a. $\frac{11}{28}$

b. $\frac{3}{7}$

c. $\frac{1}{4}$

d. $\frac{5}{14}$ ✓

Total marbles $= 28$. $P(\text{green}) = \frac{10}{28} = \frac{5}{14}$.

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30 [Intermediate] • Target: Measures of center

A jar has 7 yellow, 1 green, and 9 red marbles. Find $P(\text{yellow})$.

a. $\frac{7}{17}$ ✓ **b.** $\frac{9}{17}$

c. $\frac{1}{17}$ **d.** $\frac{10}{17}$

Total marbles $= 17$. $P(\text{yellow}) = \frac{7}{17}$.

31 [Intermediate] • Target: Measures of spread

A jar has 10 red, 8 purple, and 9 blue marbles. Find $P(\text{red})$.

a. $\frac{1}{3}$ **b.** $\frac{8}{27}$

c. $\frac{10}{27}$ ✓ **d.** $\frac{4}{9}$

Total marbles $= 27$. $P(\text{red}) = \frac{10}{27}$.

32 [Intermediate] • Target: Basic probability

A jar has 5 yellow, 9 orange, and 3 blue marbles. Find $P(\text{orange})$.

a. $\frac{9}{17}$ ✓ **b.** $\frac{5}{17}$

c. $\frac{11}{17}$ **d.** $\frac{3}{17}$

Total marbles $= 17$. $P(\text{orange}) = \frac{9}{17}$.

33 [Intermediate] • Target: Combined probability: AND / OR

A jar has 12 green, 12 orange, and 2 blue marbles. Find $P(\text{blue})$.

a. $\frac{5}{26}$ **b.** $\frac{1}{13}$ ✓

c. $\frac{2}{13}$ **d.** $\frac{6}{13}$

Total marbles $= 26$. $P(\text{blue}) = \frac{2}{26} = \frac{1}{13}$.

34 [Intermediate] • Target: Probability without replacement

A jar has 7 red, 1 yellow, and 9 green marbles. Find $P(\text{red})$.

a. $\frac{10}{17}$ **b.** $\frac{7}{17}$ ✓

c. $\frac{9}{17}$ **d.** $\frac{1}{17}$

Total marbles $= 17$. $P(\text{red}) = \frac{7}{17}$.

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35 [Intermediate] • Target: Data displays

A jar has 9 blue, 10 orange, and 3 red marbles. Find $P(\text{blue})$.

a. $\frac{5}{11}$

b. $\frac{9}{22}$ ✓

c. $\frac{3}{22}$

d. $\frac{1}{2}$

Total marbles $= 22$. $P(\text{blue}) = \frac{9}{22}$.

36 [Intermediate] • Target: Expected value

A data set has values 32, 22, 25, 31, 15, 37 plus one outlier, 118. By how much does the mean change (increase) when the outlier is included, compared to without it?

a. 40

b. 27

c. 14

d. **13** ✓

mean without outlier $= 27$, mean with outlier $= 40$, diff $= 13$

37 [Intermediate] • Target: Effects of transformations on statistics

A researcher is studying the relationship between the number of hours spent studying for a statistics exam and the score achieved. A random sample of 10 students yielded the following data, where x is the number of hours studied and y is the exam score: | Hours Studied (x) | Exam Score (y) | |---|---| | 2 | 65 | | 3 | 70 | | 4 | 78 | | 5 | 82 | | 6 | 88 | | 7 | 91 | | 8 | 95 | | 9 | 98 | | 10 | 100 | | 11 | 103 | The researcher models the relationship using a linear regression line of the form $y = \hat{a}x + \hat{b}$. Calculate the predicted exam score for a student who studies for $x = 7.5$ hours, given that the calculated regression line is $y = 4.5x + 55$.

a. **97.5** ✓

b. 95.0

c. 100.0

d. 92.5

The problem provides a linear regression model $\hat{y} = 5x + 60$, where x is the number of hours studied and $\hat{y}$ is the predicted exam score. We are asked to find the predicted exam score for a student who studies for $x = 7.5$ hours. To find the predicted exam score, we substitute $x = 7.5$ into the regression equation: $\hat{y} = 5(7.5) + 60$ First, calculate the product of 5 and 7.5: $5 \times 7.5 = 37.5$ Now, add 60 to this result: $\hat{y} = 37.5 + 60$ $\hat{y} = 97.5$ Therefore, the predicted exam s.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: a survey of students • Cross-Lesson: 1.01 Heart of Algebra

In a survey of 100 students, 60 like math, 50 like physics, and 30 like both. If a student is selected at random, what is the probability that the student likes math OR physics?

a. $\frac{4}{5}$ ✓

b. $\frac{7}{10}$

c. $\frac{3}{10}$

d. $\frac{9}{10}$

$P(M \cup P) = P(M) + P(P) - P(M \cap P) = (60 + 50 - 30)/100 = 80/100 = 4/5$.

39 [Challenging] • Target: Measures of center • Cross-Lesson: 2.01 Quadratics & Polynomials

In a survey of 40 secondary students, 18 favor Math, 14 favor Science, and 6 favor both. If a secondary student is chosen at random, what is the probability that they favor Math or Science?

a. 50%

b. 55%

c. 60%

d. **65%** ✓

$P(\text{Math} \cup \text{Science}) = \frac{18 + 14 - 6}{40} = \frac{26}{40} = 65\%$.

40 [Challenging] • Target: Measures of spread • Cross-Lesson: 3.01 Functions & Graphs

The table shows 5 students' exam scores. What is the average (arithmetic mean) of the data?

a. 82

b. 84

c. **86** ✓

d. 88

$\frac{82 + 91 + 76 + 88 + 93}{5} = \frac{430}{5} = 86$.

41 [Challenging] • Target: Basic probability • Cross-Lesson: 4.01 Ratios, Rates & Data

What is the range of the data set: 5, 21, 44, 77, 3?

a. 77

b. 73

c. **74** ✓

d. 75

Applying Statistics & Probability principles, $\text{range} = 77 - 3 = 74$ Therefore, the verified correct option is 74.

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42 [Challenging] • Target: Combined probability: AND / OR • Cross-Lesson: 6.01 Geometry

What is the median of the data set: 60, 63, 9, 35, 14?

a. 9

b. **35** ✓

c. 63

d. 36

Sorted : $9, 14, 35, 60, 63 \Rightarrow \text{median} = 35$

43 [Challenging] • Target: Probability without replacement • Cross-Lesson: 7.01 Trigonometry & Additional Topics

What is the mean (average) of the data set: 23, 24, 43, 26?

a. 28

b. 30

c. 26

d. **29** ✓

$\text{mean} = \frac{23 + 24 + 43 + 26}{4} = 29$

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44 [Challenging] • Target: Data displays • Cross-Lesson: 8.01 Exponentials & Logarithms

A quality control process at a manufacturing plant involves sampling widgets from two production lines, Line A and Line B. The number of defective widgets per batch of 100 is recorded. The data for the last 10 batches from each line is presented in the two-way table below. After observing this data, an engineer decides to implement a new quality improvement technique on Line B. They hypothesize that after the implementation, the mean number of defects per batch from Line B will decrease, and the variance of defects per batch will also decrease. To test this, they collect a new sample of 10 batches from Line B after the improvement. The number of defects in these new batches are: 5, 3, 6, 4, 5, 2, 7, 4, 5, 3. Initially, the production data was as follows: | Line | Batch 1 | Batch 2 | Batch 3 | Batch 4 | Batch 5 | Batch 6 | Batch 7 | Batch 8 | Batch 9 | Batch 10 |
|---|---|---|---|---|---|---|---|---|---|---| | Line A | 8 | 7 | 9 | 8 | 10 | 7 | 8 | 9 | 8 | 9 | | Line B | 12 | 10 | 11 | 13 | 10 | 12 | 11 | 14 | 10 | 12 | Calculate the sample variance for the original Line B data and the sample variance for the new Line B data. Based on these sample variances, which of the engineer's hypotheses is directly supported by the calculated values, assuming an alpha level of 0.05 for any formal statistical test (though no formal test is required here, only comparison of sample variances)?

a. **The mean number of defects decreased, and the variance decreased.** ✓

b. The mean number of defects increased, and the variance decreased.

c. The mean number of defects decreased, and the variance increased.

d. The mean number of defects increased, and the variance increased.

First, let's calculate the sample mean and sample variance for the original Line B data. The original Line B defects are: 12, 10, 11, 13, 10, 12, 11, 14, 10, 12. Number of batches, n_{old} = 10. Sum of defects = 12+10+11+13+10+12+11+14+10+12 = 115. Sample mean for original Line B, $\bar{x}_{old} = \frac{115}{10}$ = 11.5. To calculate the sample variance, we need the sum of squared differences from the mean. Sum of $(x_i - \bar{x}_{old})^2$.

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45 [Challenging] • Target: Expected value • Cross-Lesson: 1.01 Heart of Algebra

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students who chose Soccer or Basketball as their favorite sport?

a. 30

b. **40** ✓

c. 50

d. 60

The bar chart shows the number of students who chose each sport. \\ - Soccer: 22 students \\ - Basketball: 26 students \\ To find the total number of students who chose Soccer or Basketball, we add the number of students for each sport: \\ $22 + 26 = 48$ \\ However, examining the choices, there might be a slight discrepancy or the question might be testing simple addition from the chart directly. Let's re-examine the bar chart data carefully. \\ If we assume the labels and data are as shown in the provided bar char.

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